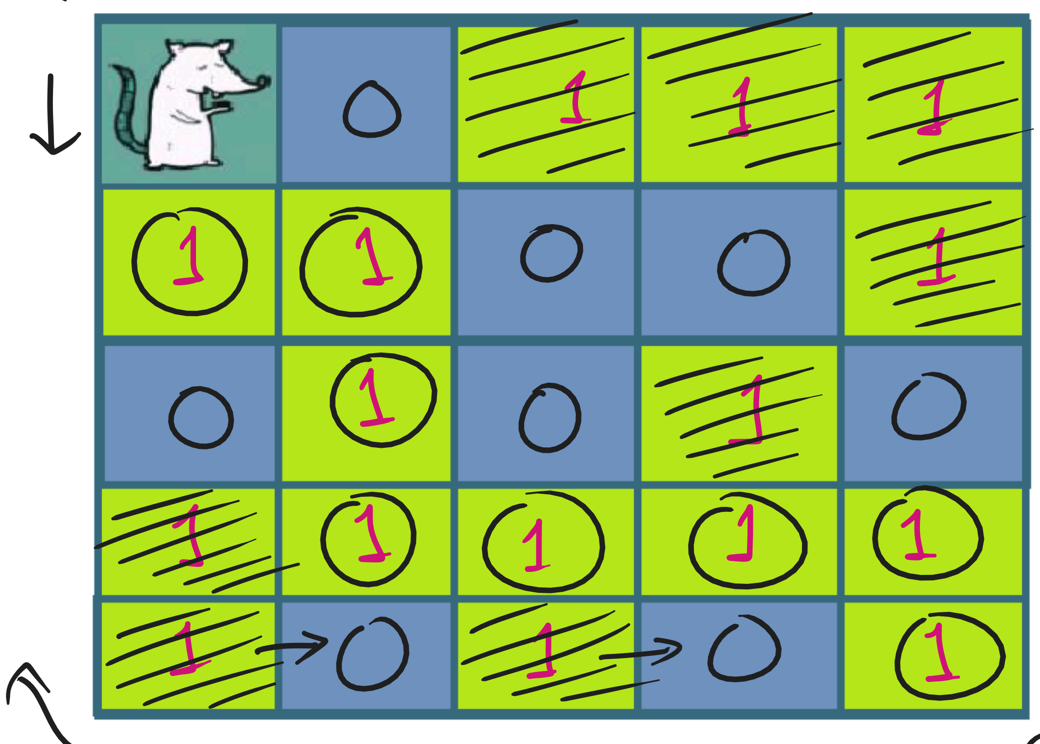
Applications of Recursion :→ Back Tracking

x y
(0, 0) →   ☐ safe 1, 0   ~~1 0~~ 0 1 0 ← wall



Back tracking
Recursion

Rat In A Maze :→

Forward : $(x+1, y)$

Downward : $(x, y+1)$

0 → Walls
1 → Path

destination
Food

x y
$(n-1, n-1)$
5, 5
4, 4

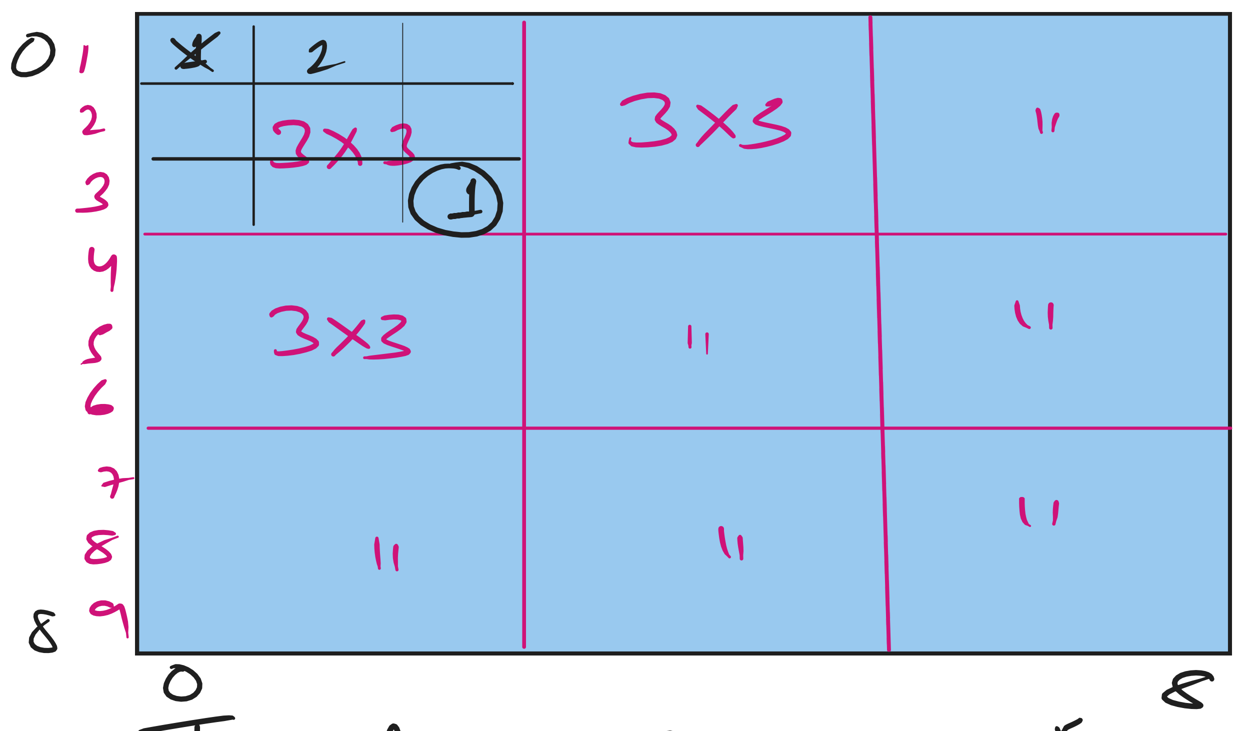
* Rat cannot hit the wall.
* Can travel only via Green

* N Queens [ ]
* Rat In A Maze
* Sudoku Solver
* Subsets of A String

* Greedy Algos
* Dynamic Programming → Tabulation, Recursion, Memoisation, Space Optimization
* Bit Masking

Backtracking Sudoku Solver : 9×9 Matrix

(0, 0)



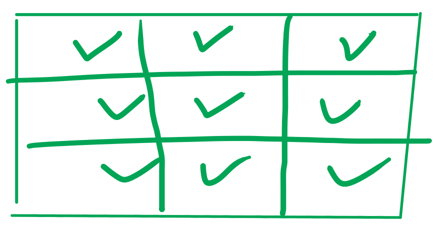
1. Each Row
2. Each Column
3. Each 3×3 Matrix

* Should have (1-9) only once.

Target → 3×3 → Insert 1 to 9 once :

1. isPathSafe(x, y) (row)(col) == val

Formula for row & column [cell] in a 3×3 Matrix :→



Matrix [row] [col]

$[3*(row/3)+i/3]\ [3\times(col/3)+i\%3]$

$i=0,\ r=0,\ c=0$

$[3\times 0/3+i/3]$

$[0+0]\quad [3\times 0+0]$

$[0]\quad [0]$